# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: Computer Vision with OpenCV**

**COURSE CODE: DSA0202**

**COURSE OUTCOME COVERED**

***CO2:*** *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

*Assessment Tool Weightage: 50%*

**QUESTIONS: 5 Total Marks:50** Annexure A

## Analytical Problem Solving

***Code of Conduct:***

*I K . Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary*

*action.* ***Signature of the student: K. Pujitha***

RUBRICS FOR CALCULATION

**Numerical Problems**

| **Criteria** | **Weightage** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** | **Marks** |
|---|---|---|---|---|---|---|
| Problem Underst and ing | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem | |
| Method Selection | 20% | Selects most appropriate method/formula with | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | strong justification | | | | |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process | |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations | |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation | |

Annexure A

# Analytical Problem Solving

**Question 1: Salt-and-Pepper Noise Removal Before Edge Detection** Scenario: A grayscale image of size 1024 × 1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

a. Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.

b. Explain why the selected filter performs better than an averaging filter for impulse noise removal.

c. If a 3 × 3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.

d. Estimate the total number of noisy pixels present in the original image. e. Estimate the expected number of noisy pixels remaining after median filtering. f. Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

**Question 2: Discretization of a Continuous Image**

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x, while it remains constant along the y-direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

a. Explain how the continuous image can be converted into a digital image using sampling and quantization.

b. Identify the spatial variation pattern of the image in the x-direction. c. Determine the minimum sampling requirement needed to avoid aliasing in the x direction.

d. Explain what will happen if the image is sampled below the required sampling rate. e. Suggest a suitable 8-bit quantization scheme for storing the image. f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

**Question 3: Edge Detection in Thin Line Images**

Scenario: A 512 × 512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answer the following:

a. Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.

b. For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.

c. Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.

d. Discuss the role of Gaussian smoothing in the Canny edge detection process. e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.

f. If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

**Question 4: Feature Extraction for Handwritten Digit Recognition** Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification.

Answer the following:

a. Explain the working principles of SIFT, SURF, and ORB feature extraction

methods. b. Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.
c. If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.
d. If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.
e. Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.
f. Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

**Question 5: Morphological Preprocessing for Character Recognition** Scenario: A binary image of size 256 × 256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.
Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.
Answer the following:
a. Identify the morphological operation suitable for removing small white noise dots from the background.
b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.
c. Explain the difference between opening and closing operations using simple image processing terminology.
d. If a 3 × 3 square structuring element is used, explain how it affects thin character strokes.
e. Discuss the risk of using a large structuring element during preprocessing. f. Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

## 1. Given Data

- Image size = 1024 × 1024 pixels
- Noise type = Salt-and-Pepper (Impulse) Noise
- Noise density = 0.05 (5%)
- The image is to be preprocessed before applying an edge detection algorithm.

**(a) Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.**

**Answer**

The most suitable filtering technique is the Median Filter, preferably a 3 × 3 Median Filter.

Salt-and-pepper noise appears as randomly occurring black (0 intensity) and white (255 intensity) pixels in an image. Since this type of noise affects only a small number of pixels, the median filter is highly effective in removing it.

Unlike linear filters, the median filter replaces the center pixel with the median value of all the pixels in the selected neighborhood. Because the median is not affected by extreme values, the noisy pixels are removed while the original image details are preserved.

The median filter also preserves edges, which is very important before performing edge detection because blurred edges reduce the accuracy of edge detection algorithms.

Therefore, a 3 × 3 Median Filter is the most suitable preprocessing technique for this image.

**(b) Explain why the selected filter performs better than an averaging filter for impulse noise removal.**

**Answer**

The median filter performs much better than an averaging filter for removing salt-and-pepper noise because of the way each filter processes pixel values.

The averaging filter calculates the average of all pixels in the neighborhood. If one or two pixels have very high values (255) or very low values (0), these extreme values affect the average and produce blurred results.

For example,

Consider the following 3 × 3 neighborhood:

20 21 19
22 255 20
21 19 20

Here, 255 represents a noisy pixel.

**Using Averaging Filter**

Average value

= (20 + 21 + 19 + 22 + 255 + 20 + 21 + 19 + 20) / 9

= 397 / 9

≈ 44

Thus, the noisy value influences the average, and the output becomes 44, which is not close to the original pixel value.

**Using Median Filter**

Arrange the values in ascending order.

19
20
20
20
21
21
22
255

The middle value (median) is 20.

Hence, the noisy pixel is replaced with 20, which is very close to the original image intensity.

**Advantages of Median Filter**

- Removes impulse (salt-and-pepper) noise effectively.
- Preserves image edges.
- Does not blur important details.
- Is not influenced by extreme pixel values.
- Produces better image quality before edge detection.

Therefore, the median filter is preferred over the averaging filter for salt-and-pepper noise removal.

**(c) If a 3 × 3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.**

**Answer**

**Assumptions**

1. Noise is randomly distributed.
2. Noise density is 5% (0.05).
3. A noisy center pixel will be corrected if the majority of the surrounding pixels are not noisy.
4. The neighboring pixels are independent of one another.

The probability that one neighboring pixel is not noisy is

P=1−0.05=0.95P = 1 - 0.05 = 0.95P=1−0.05=0.95

A 3 × 3 window contains 8 neighboring pixels around the center pixel.

The probability that all eight neighboring pixels are noise-free is

P=(0.95)8P=(0.95)^8P=(0.95)8 P=0.6634P=0.6634P=0.6634

Therefore,

P≈66.3%P \approx 66.3\%P≈66.3%

**Result**

The probability that a noisy center pixel will be corrected is approximately 66.3% under the above assumptions.

**Note:** In practical situations, the correction rate is often higher because the median filter only requires most neighboring pixels to be noise-free, not necessarily all of them.

**(d) Estimate the total number of noisy pixels present in the original image.**

**Answer**

Image dimensions

1024×10241024 \times 10241024×1024

Total number of pixels

=1,048,576=1,048,576=1,048,576

Noise density

=5%=0.05=5\%=0.05=5%=0.05

Total noisy pixels

=1,048,576×0.05=1,048,576 \times 0.05=1,048,576×0.05
=52,428.8=52,428.8=52,428.8

Approximately,

52,429 noisy pixels\boxed{52,429\text{ noisy pixels}}52,429 noisy pixels

Thus, about 52,429 pixels in the image are affected by salt-and-pepper noise.

**(e) Estimate the expected number of noisy pixels remaining after median filtering.**

**Answer**

From part (d),

Total noisy pixels

=52,429=52,429=52,429

From part (c),

Correction probability

=66.3%=0.663=66.3\%=0.663=66.3%=0.663

Remaining noisy pixels

=52,429×(1−0.663)=52,429 \times (1-0.663)=52,429×(1−0.663)
=52,429×0.337=52,429 \times 0.337=52,429×0.337 =17,668=17,668=17,668

Therefore,

Remaining noisy pixels≈17,668\boxed{\text{Remaining noisy pixels} \approx 17,668}Remaining noisy pixels≈17,668

This is only an estimated value based on the assumptions made in part (c). In real applications, the median filter generally removes a larger percentage of salt-and-pepper noise.

**(f) Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.**

**Answer**

If the image is affected by Gaussian noise, the preprocessing method should be changed because Gaussian noise behaves differently from salt-and-pepper noise.

Salt-and-pepper noise appears as isolated black and white pixels, whereas Gaussian noise introduces small random intensity variations over the entire image.

Since Gaussian noise affects almost every pixel, the median filter is not the best choice.

Instead, filters such as the Gaussian Filter, Mean (Averaging) Filter, or Wiener Filter are preferred.

The Gaussian filter smooths the image by assigning higher importance to nearby pixels and lower importance to distant pixels using a Gaussian distribution. It effectively reduces Gaussian noise while preserving the overall image structure.

The Wiener filter performs even better because it adapts according to the local statistics of the image and provides a good balance between noise removal and edge preservation.

Therefore, when Gaussian noise is present, the preprocessing stage should use a Gaussian Filter or Wiener Filter instead of a median filter before applying edge detection.

## 2) Given Data

- The image is an analog grayscale image.
- The intensity varies as a smooth sinusoidal wave along the x-direction.
- The intensity remains constant along the y-direction.
- The spatial range of the image is 0 to 255 in both x and y directions.
- The image must be converted into a digital image for storage and processing.

**(a) Explain how the continuous image can be converted into a digital image using sampling and quantization.**

**Answer**

A continuous image cannot be directly processed by a computer because computers can only handle digital data. Therefore, the analog image must be converted into a digital image through sampling and quantization.

**1. Sampling**

Sampling is the process of converting the continuous spatial coordinates of an image into discrete pixel locations.

- The analog image contains infinite intensity values at every point.
- During sampling, the image is measured at regular intervals along the x and y directions.
- Each sampled point becomes one pixel in the digital image.
- The number of samples determines the spatial resolution of the image.

Higher sampling produces more pixels and better image detail, while lower sampling reduces image quality.

**2. Quantization**

After sampling, each sampled pixel still has a continuous intensity value.

Quantization converts these continuous intensity values into a finite number of discrete gray levels.

For an 8-bit image,

28=2562^8 = 25628=256

gray levels are available.

The intensity values are represented between

- **0 = Black**
- **255 = White**

Each sampled pixel is assigned to one of these 256 gray levels.

**Digital Image Formation**

Therefore, digital image conversion takes place in two stages:

- **Sampling** → Converts continuous spatial coordinates into pixels.
- **Quantization** → Converts continuous intensity values into discrete gray levels.

After these two steps, the analog image becomes a digital image that can be stored, displayed, and processed by a computer.

**(b) Identify the spatial variation pattern of the image in the x-direction.**

**Answer**

The image intensity varies according to a sinusoidal pattern along the x-direction.

This means that:

- The brightness increases gradually.
- It reaches a maximum value.
- It then decreases gradually.
- It reaches a minimum value.
- The same pattern repeats periodically.

Mathematically, the intensity can be represented as

I(x)=Asin(2πfx)+BI(x)=A\sin(2\pi fx)+BI(x)=Asin(2πfx)+B

where

- **A** = Amplitude
- **f** = Spatial frequency

- **B** = Average intensity

Since the intensity does not change along the y-direction, every horizontal row has the same sinusoidal variation.

Thus, the image consists of repeating bright and dark vertical bands.

**(c) Determine the minimum sampling requirement needed to avoid aliasing in the x-direction.**

**Answer**

According to the Nyquist Sampling Theorem, a signal must be sampled at least twice its highest frequency to reconstruct it correctly.

If

$$f=\text{highest spatial frequency}$$

then the minimum sampling frequency is

$$f_s \ge 2f$$

This is known as the Nyquist Rate.

Therefore, the image should be sampled at at least twice the highest spatial frequency present in the sinusoidal image.

Sampling at this rate preserves the original sinusoidal pattern without distortion.

**(d) Explain what will happen if the image is sampled below the required sampling rate.**

**Answer**

If the image is sampled below the Nyquist sampling rate**,** aliasing occurs.

Aliasing means that the original high-frequency pattern is incorrectly represented as a lower-frequency pattern.

As a result,

- Fine image details are lost.
- False image patterns appear.
- The sinusoidal wave appears distorted.
- Jagged edges may appear.
- Image reconstruction becomes impossible.

The computer interprets the original pattern incorrectly because there are not enough samples to represent the actual image.

Therefore, undersampling reduces image quality and produces inaccurate results during image processing.

**(e) Suggest a suitable 8-bit quantization scheme for storing the image.**

**Answer**

An 8-bit uniform quantization scheme is suitable for storing the image.

In an 8-bit grayscale image,

28=2562^8=25628=256

gray levels are available.

The intensity range is

0 to 2550 \text{ to }2550 to 255

Each pixel is represented using 8 bits (1 byte).

The gray levels are

- 0 → Black
- 1–254 → Different shades of gray
- 255 → White

Uniform quantization divides the entire intensity range into equal intervals, making it simple, efficient, and widely used in digital image processing.

Therefore, an 8-bit uniform quantization scheme is the most suitable choice.

**(f) Discuss the effect of quantization on image quality, storage size, and intensity accuracy.**

**Answer**

Quantization affects several important characteristics of a digital image.

**1. Effect on Image Quality**

Higher bit depth provides smoother intensity transitions and better visual quality.

Lower bit depth causes visible intensity jumps, false contours, and loss of image details.

Therefore,

Higher bit depth → Better image quality.

**2. Effect on Storage Size**

The storage requirement depends on the number of bits used per pixel.

For an image of size

1024×10241024 \times 10241024×1024

using 8-bit quantization,

Storage required

1024×1024×8=8,388,608 bits1024 \times 1024 \times 8 =8,388,608 \text{ bits}1024×1024×8=8,388,608 bits

=8,388,6088=1,048,576 bytes=\frac{8,388,608}{8} =1,048,576 \text{ bytes}=88,388,608=1,048,576 bytes

Approximately,

1 MB1\text{ MB}1 MB

Thus,

Higher bit depth increases storage requirements.

### 3. Effect on Intensity Accuracy

Higher quantization levels provide a more accurate representation of the original image.

Lower quantization levels introduce quantization error, where the original intensity values are rounded to the nearest available gray level.

This reduces intensity accuracy and may produce visible artifacts.

Therefore,

- More gray levels → Higher intensity accuracy.
- Fewer gray levels → Larger quantization error.

## 3) Given Data

- Image size = 512 × 512 pixels
- Image type = Grayscale
- Contains 1-pixel-thick vertical and horizontal white lines
- Background = Uniform dark (black)
- Objective = Detect line boundaries accurately without losing thin edge information.

**(a) Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.**

## Answer

Edge detection is an important image processing technique used to identify object boundaries by detecting sudden changes in pixel intensity. Three commonly used edge detection methods are **Sobel**, **Prewitt**, and **Canny**.

### 1. Sobel Edge Detector

The Sobel operator is a first-order gradient-based edge detector.

- Uses **3 × 3 convolution masks**.
- Gives more weight to the center pixels.
- Calculates gradients in both horizontal and vertical directions.
- Less sensitive to noise than the Prewitt operator.
- Produces moderately accurate edges.

#### Advantages

- Simple and fast.
- Detects horizontal and vertical edges effectively.
- Slightly smooths noise while computing gradients.

#### Disadvantages

- Thick edges may be produced.
- Very thin (1-pixel) lines may not always be preserved accurately.

### 2. Prewitt Edge Detector

The Prewitt operator is also a gradient-based edge detector.

- Uses equal weights in the convolution mask.
- Simpler than Sobel.
- Faster computation.

### Advantages

- Easy to implement.
- Suitable for simple images with low noise.

### Disadvantages

- More sensitive to noise.
- Produces weaker edge responses.
- Less accurate than Sobel.

## 3. Canny Edge Detector

The Canny operator is considered one of the best edge detection techniques.

It consists of several stages:

1. Gaussian smoothing
2. Gradient calculation
3. Non-maximum suppression
4. Double thresholding
5. Edge tracking by hysteresis

### Advantages

- Produces thin and continuous edges.
- Excellent noise removal.
- Very accurate edge localization.
- Preserves 1-pixel-wide edges effectively.

### Disadvantages

- More computationally expensive.
- Requires selection of threshold values.

## Comparison Table

| Feature | Sobel | Prewitt | Canny |
|---|---|---|---|
| Noise immunity | Moderate | Low | High |
| Edge accuracy | Good | Moderate | Excellent |
| Produces thin edges | No | No | Yes |
| Computational complexity | Low | Low | High |
| Suitable for 1-pixel lines | Moderate | Poor | Excellent |

**(b) For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.**

**Answer**

The Sobel operator uses two 3 × 3 kernels.

**Horizontal Gradient (Gx)**

[−101−202−101]\begin{bmatrix} -1&0&1\\ -2&0&2\\ -1&0&1 \end{bmatrix}
−1−2−1000121

**Vertical Gradient (Gy)**

[−1−2−1000121]\begin{bmatrix} -1&-2&-1\\ 0&0&0\\ 1&2&1 \end{bmatrix}
−101−202−101

Assume the horizontal edge has the following pixel values:

```
255    255    255
255    255    255
0        0        0
```

where

- 255 = White
- 0 = Black

**Step 1: Calculate Horizontal Gradient (Gx)**

Gx=0G_x=0Gx=0

because there is no intensity variation in the horizontal direction.

**Step 2: Calculate Vertical Gradient (Gy)**

=(−1×255)+(−2×255)+(−1×255)+(0×255)+(0×255)+(0×255)+(1×0)+(2×0)+(1×0)=(-1×255)+(-2×255)+(-1×255) +(0×255)+(0×255)+(0×255) +(1×0)+(2×0)+(1×0)=(−1×255)+(−2×255)+(−1×255)+(0×255)+(0×255)+(0×255)+(1×0)+(2×0)+(1×0) =−255−510−255=-255-510-255=−255−510−255 =−1020=-1020=−1020

Magnitude

∣Gy∣=1020|G_y|=1020∣Gy∣=1020

**Step 3: Gradient Magnitude**

G=Gx2+Gy2G=\sqrt{G_x^2+G_y^2}G=Gx2+Gy2 =02+10202=\sqrt{0^2+1020^2}=02+10202 =1020=1020=1020

**Result**

- Horizontal Gradient

Gx=0G_x=0Gx=0

- Vertical Gradient

Gy=−1020G_y=-1020Gy=−1020

- Gradient Magnitude

G=1020G=1020G=1020

This indicates a strong horizontal edge.

**(c) Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.**

**Answer**

The gradient represents the rate of change in image intensity.

It has two important components.

**1. Gradient Magnitude**

Gradient magnitude indicates how strong an edge is.

It is calculated as

G=Gx2+Gy2G=\sqrt{G_x^2+G_y^2}G=Gx2+Gy2

A larger gradient magnitude means

- Strong edge
- Sudden intensity change

A smaller gradient magnitude means

- Weak edge
- Smooth intensity variation

**2. Gradient Direction**

Gradient direction indicates the orientation of the edge.

It is calculated using

θ=tan−1(GyGx)\theta=\tan^{-1}\left(\frac{G_y}{G_x}\right)θ=tan−1(GxGy)

The direction tells whether the edge is

- Horizontal

- Vertical
- Diagonal

This information is used during Non-Maximum Suppression in the Canny algorithm to produce thin edges.

Therefore,

- Gradient magnitude determines edge strength.
- Gradient direction determines edge orientation.

**(d) Discuss the role of Gaussian smoothing in the Canny edge detection process.**

**Answer**

Before detecting edges, the Canny algorithm first applies Gaussian smoothing.

Its purpose is to remove random image noise that may produce false edges.

The Gaussian filter replaces each pixel with a weighted average of neighboring pixels.

The center pixels receive larger weights than distant pixels.

The benefits of Gaussian smoothing are:

- Removes high-frequency noise.
- Reduces false edge detection.
- Produces smoother gradient calculations.
- Improves edge detection accuracy.
- Increases reliability of Canny edge detection.

Without Gaussian smoothing, even small noise variations may be detected as edges.

Therefore, Gaussian smoothing is an essential preprocessing step in the Canny algorithm.

**(e) Explain how Gaussian smoothing may affect edge localization, especially for thin lines.**

**Answer**

Although Gaussian smoothing removes noise effectively, it also has some disadvantages.

When the filter smooths the image,

- Edge sharpness decreases.
- Thin edges become wider.
- Very fine details may disappear.
- Exact edge position may shift slightly.

For 1-pixel-thick lines, excessive smoothing may

- Blur the line.
- Reduce edge strength.
- Merge nearby edges.
- Cause loss of very thin structures.

Therefore, a small Gaussian kernel should be selected for thin-line images so that noise is reduced without significantly affecting edge localization.

**(f) If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.**

**Answer**

The Canny algorithm uses two thresholds:

- High threshold
- Low threshold

These thresholds decide which gradients are considered edges.

If the threshold is reduced to half,

**Detected Edges**

More edges will be detected because weaker gradients also satisfy the threshold condition.

**Weak Edges**

Many weak edges that were previously ignored will now appear in the output image.

Some of these may represent genuine image details.

**False Edges**

Noise also produces weak gradients.

Since the threshold is lower,

many noisy pixels will also be detected as edges.

This increases the number of false edges.

**Overall Effect**

Advantages:

- More complete edge detection.
- Better detection of faint lines.

Disadvantages:

- Increased noise.
- More false edges.
- Reduced edge quality.

Therefore, selecting an appropriate threshold is important to achieve a balance between detecting real edges and avoiding false edge detection.

## 4) Given Data

- A computer vision system is developed for handwritten digit recognition.
- The digit images are obtained from scanned documents.
- The handwritten digits vary in:
  - Size
  - Orientation
  - Stroke thickness
  - Writing style
- Robust feature extraction is required before classification.

**(a) Explain the working principles of SIFT, SURF, and ORB feature extraction methods.**

**Answer**

Feature extraction is the process of identifying distinctive points or patterns in an image that can be used to recognize and match objects. In handwritten digit recognition, feature extraction helps identify important characteristics of each digit despite variations in writing style, size, or orientation.

The three widely used feature extraction methods are SIFT (Scale-Invariant Feature Transform), SURF (Speeded-Up Robust Features), and ORB (Oriented FAST and Rotated BRIEF).

### 1. SIFT (Scale-Invariant Feature Transform)

SIFT is one of the most accurate feature extraction techniques used in computer vision.

**Working Principle**

The SIFT algorithm works in four stages:

**Step 1: Scale-space Extrema Detection**

- The image is blurred using Gaussian filters at different scales.
- Difference of Gaussian (DoG) images are created.
- Keypoints are detected by identifying maxima and minima across different scales.

**Step 2: Keypoint Localization**

- Weak or unstable keypoints are removed.
- Only stable and meaningful keypoints are retained.

**Step 3: Orientation Assignment**

- Each keypoint is assigned a dominant orientation based on the local image gradient.
- This provides rotation invariance.

**Step 4: Keypoint Descriptor Generation**

- A 128-dimensional feature descriptor is generated for every keypoint.
- These descriptors are used for matching similar features between images.

**Advantages**

- Highly accurate.
- Scale invariant.
- Rotation invariant.
- Robust against illumination changes.

**Disadvantages**

- High computational cost.
- Slower execution.
- Not suitable for real-time applications.

**2. SURF (Speeded-Up Robust Features)**

SURF is an improved version of SIFT designed to increase processing speed.

### Working Principle

The SURF algorithm includes the following steps:

#### Step 1: Interest Point Detection

- Uses the Hessian Matrix to detect keypoints.
- Integral images are used to speed up computations.

#### Step 2: Orientation Assignment

- Wavelet responses determine the dominant orientation.

#### Step 3: Feature Descriptor

- A 64-dimensional descriptor is generated.
- This descriptor is shorter than SIFT, making SURF faster.

### Advantages

- Faster than SIFT.
- Scale invariant.
- Rotation invariant.
- Good matching accuracy.

### Disadvantages

- Slightly less accurate than SIFT.
- Still computationally expensive for some real-time systems.

## 3. ORB (Oriented FAST and Rotated BRIEF)

ORB is designed for fast feature extraction in real-time computer vision applications.

### Working Principle

ORB combines two algorithms:

- FAST (Features from Accelerated Segment Test) for keypoint detection.

- BRIEF (Binary Robust Independent Elementary Features) for feature description.

The steps are:

**Step 1: FAST Corner Detection**

- Detects corners efficiently.

**Step 2: Orientation Assignment**

- Calculates the orientation of each keypoint.

**Step 3: Rotated BRIEF Descriptor**

- Generates a binary feature descriptor.
- The descriptor is rotated according to the keypoint orientation.

**Advantages**

- Very fast.
- Low memory usage.
- Rotation invariant.
- Suitable for real-time applications.

**Disadvantages**

- Less accurate than SIFT and SURF.
- Less robust to significant scale changes.

**(b) Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.**

**Answer**

| Feature | SIFT | SURF | ORB |
|---|---|---|---|
| Scale Invariance | Excellent | Excellent | Moderate |
| Rotation Invariance | Excellent | Excellent | Good |
| Computational Cost | High | Medium | Low |

| Feature | SIFT | SURF | ORB |
|---|---|---|---|
| Processing Speed | Slow | Fast | Very Fast |
| Descriptor Size | 128 | 64 | Binary Descriptor |
| Real-Time Suitability | No | Limited | Yes |
| Matching Accuracy | Very High | High | Moderate |
| Memory Requirement | High | Medium | Low |

**Conclusion**

- **SIFT** provides the highest matching accuracy but requires more computation.
- **SURF** offers a balance between speed and accuracy.
- **ORB** is the best choice for real-time handwritten digit recognition due to its fast processing speed and low computational cost.

**(c) If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.**

**Answer**

A rotation of **45°** changes the orientation of the handwritten digit.

The feature extraction algorithm should therefore be rotation invariant.

- **SIFT** assigns an orientation to every keypoint, allowing it to match features accurately even after rotation.
- **SURF** also assigns orientations using wavelet responses and can reliably match rotated images.
- **ORB** estimates keypoint orientation and rotates the BRIEF descriptor, enabling reliable matching for rotated images.

Therefore, all three methods (**SIFT, SURF, and ORB**) can produce reliable matching features when the digit image is rotated by 45°.

However,

- **SIFT** provides the highest matching accuracy.
- **SURF** performs well with faster execution.

- **ORB** is suitable for real-time applications but may be slightly less accurate under large scale variations.

**(d) If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.**

**Answer**

Average keypoints per image

=120=120=120

Number of images

=50=50=50

Total keypoints

=120×50=120\times50=120×50 =6000=6000=6000

**Result**

Total keypoints detected=6000\boxed{\text{Total keypoints detected}=6000}Total keypoints detected=6000

Therefore, 6000 keypoints are detected from the 50 handwritten digit images.

**(e) Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.**

**Answer**

The most suitable dimensionality reduction technique is Principal Component Analysis (PCA).

**Working Principle of PCA**

PCA transforms the original high-dimensional feature vectors into a smaller number of new variables called Principal Components.

These principal components:

- Capture the maximum variance present in the data.
- Remove redundant information.
- Reduce feature dimensionality while preserving most of the useful information.

To retain 95% of the useful information, the first few principal components whose cumulative variance reaches 95% are selected.

This significantly reduces the feature size while maintaining almost the same recognition performance.

Therefore, Principal Component Analysis (PCA**)** is the most suitable dimensionality reduction technique.

**(f) Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.**

**Answer**

Dimensionality reduction improves the performance of handwritten digit recognition in several ways.

**1. Reduces Computational Time**

Fewer features require fewer mathematical operations.

As a result,

- Feature extraction becomes faster.
- Classification becomes faster.

**2. Reduces Memory Usage**

High-dimensional feature vectors occupy more storage.

Reducing the number of features decreases memory requirements.

### 3. Removes Redundant Features

Many extracted features contain similar information.

Dimensionality reduction removes unnecessary and correlated features.

### 4. Improves Classification Accuracy

Removing noisy and irrelevant features helps the classifier focus only on useful information.

This often increases recognition accuracy.

### 5. Reduces Overfitting

When too many features are used,

the classifier may memorize the training data instead of learning general patterns.

Reducing feature dimensions helps improve the model's ability to classify new handwritten digits correctly.

### 6. Faster Training

Machine learning algorithms require less training time because fewer features are processed.

### Overall Benefits

- Faster processing
- Lower memory usage
- Better classification accuracy
- Reduced overfitting
- Improved efficiency

## 5) Given Data

- Image size = 256 × 256 pixels
- Image type = Binary image
- Contains scanned handwritten characters
- Due to poor scanning quality:
    - Small white noise dots are present in the background.
    - Small black gaps (holes**)** are present inside the character strokes.
- The image must be preprocessed before applying a character recognition algorithm.

**(a) Identify the morphological operation suitable for removing small white noise dots from the background.**

**Answer**

The most suitable morphological operation for removing small white noise dots from the background is Opening.

**Opening Operation**

Opening is performed in two steps:

1. **Erosion**
2. **Dilation**

Mathematically,

Opening = Erosion followed by Dilation

**Working**

- During erosion, small white noise pixels that are not connected to the main object are removed.
- Dilation then restores the size of the original characters without bringing back the removed noise.

**Advantages**

- Removes isolated white noise dots.
- Preserves the overall shape of handwritten characters.
- Produces a cleaner background.
- Improves the accuracy of character recognition.

Therefore**, Opening** is the most suitable operation for removing small white noise dots.

**(b) Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.**

**Answer**

The most suitable morphological operation is Closing.

**Closing Operation**

Closing consists of two steps:

1. **Dilation**
2. **Erosion**

Mathematically,

Closing = Dilation followed by Erosion

**Working**

- Dilation expands the character strokes.
- Small holes and gaps inside the strokes are filled.
- Erosion restores the original size while keeping the filled gaps closed.

**Advantages**

- Fills small holes.
- Connects broken strokes.
- Produces continuous handwritten characters.
- Improves recognition accuracy.

Therefore, Closing is the best operation for filling small gaps inside character strokes.

**(c) Explain the difference between opening and closing operations using simple image processing terminology.**

**Answer**

Both Opening and Closing are basic morphological operations used to improve binary images, but they serve different purposes.

**Opening**

Opening is performed by applying erosion first, followed by dilation.

Its main purpose is to remove small unwanted white objects or noise from the background without significantly affecting the main object.

**Applications**

- Removes isolated white dots.
- Removes small protrusions.
- Smoothens object boundaries.
- Cleans noisy binary images.

**Closing**

Closing is performed by applying dilation first, followed by erosion.

Its main purpose is to fill small holes and gaps inside objects and connect nearby object parts.

**Applications**

- Fills small holes.
- Connects broken strokes.
- Smoothens object boundaries.
- Improves continuity of characters.

**Comparison Table**

| Opening | Closing |
|---|---|
| Erosion followed by Dilation | Dilation followed by Erosion |
| Removes small white noise | Fills small holes and gaps |
| Shrinks first, then restores | Expands first, then restores |
| Eliminates isolated objects | Connects broken objects |
| Used for noise removal | Used for gap filling |

**(d) If a 3 × 3 square structuring element is used, explain how it affects thin character strokes.**

**Answer**

A 3 × 3 square structuring element is one of the most commonly used structuring elements in morphological image processing.

Since it is relatively small, it provides a good balance between noise removal and shape preservation.

**Effects on Thin Character Strokes**

- Removes small isolated noise.
- Smoothens the character boundaries.
- Fills very small gaps between pixels.
- Preserves most handwritten strokes.

However,

very thin strokes that are only one pixel wide may become thinner during erosion.

If the strokes are extremely thin,

- Some stroke pixels may disappear.
- Small character details may be lost.
- Recognition accuracy may slightly decrease.

Despite these limitations, a $3 \times 3$ structuring element is generally considered suitable because it effectively removes noise while preserving most character information.

**(e) Discuss the risk of using a large structuring element during preprocessing.**

**Answer**

The size of the structuring element plays an important role in morphological preprocessing.

Using a large structuring element may produce several undesirable effects.

**1. Loss of Fine Details**

Small features of handwritten characters may disappear.

For example,

- Small loops
- Dots
- Curves

may be removed completely.

**2. Distortion of Character Shape**

Large structuring elements may significantly alter the original character shape.

Characters may become thicker or thinner than intended.

**3. Merging of Nearby Characters**

Characters located close together may merge into one object.

This makes segmentation difficult.

**4. Loss of Thin Strokes**

Very thin handwritten strokes may disappear completely after erosion.

**5. Reduced Recognition Accuracy**

Since character shapes are modified,

feature extraction becomes less accurate,

which reduces the performance of the character recognition system.

**Conclusion**

A structuring element should be chosen carefully. A small $3 \times 3$ structuring element is generally preferred because it removes noise while preserving important character details.

**(f) Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.**

**Answer**

A complete preprocessing pipeline for handwritten character recognition consists of the following steps:

**Step 1: Image Acquisition**

The handwritten document is scanned using a scanner or captured using a digital camera.

The output is usually a grayscale image.

**Step 2: Image Resizing**

The image is resized to a fixed dimension so that all characters have a uniform size.

This improves consistency during processing.

**Step 3: Grayscale Conversion**

If the scanned image is in RGB format, it is converted into a grayscale image.

This simplifies further processing.

**Step 4: Image Binarization**

Thresholding techniques such as Otsu's method are used to convert the grayscale image into a binary image.

Pixels become either black or white.

**Step 5: Noise Removal**

Morphological Opening removes small white noise dots.

Median filtering may also be used to remove isolated noise.

**Step 6: Gap Filling**

Morphological Closing fills small holes and broken strokes inside the handwritten characters.

**Step 7: Character Segmentation**

Each handwritten character is separated from the image.

This allows individual character processing.

**Step 8: Normalization**

Each character is resized and centered.

Stroke thickness may also be normalized.

**Step 9: Skeletonization (Optional)**

Characters are reduced to one-pixel-wide skeletons while preserving their structure.

This simplifies feature extraction.

**Step 10: Feature Extraction**

Important features are extracted using techniques such as:

- SIFT
- SURF
- ORB
- HOG (Histogram of Oriented Gradients)
- Zoning
- PCA (for dimensionality reduction)

The extracted features are used as input to the classifier.

**Step 11: Classification**

The extracted features are classified using machine learning or deep learning algorithms such as:

- Support Vector Machine (SVM)
- K-Nearest Neighbour (KNN)
- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)

The classifier predicts the handwritten character.